

SOC Investigation Report

FRONTDESK-PC1 Compromise

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Findings

Time Range: 15 October 2025, 12:51:44 UTC – 13:06:40 UTC

Host: FRONTDESK-PC1

Compromised Account: Ryan.Adams

Indicators of Compromise (IOCs)

- **Attacker IP:** 172.16.0.184
- **C2 IP:** 157.245.46.190
- **C2 Ports:** 9999 (primary), 8888 (secondary)
- **Malicious File:** C:\Users\Ryan.Adams\Music\python.exe
- **Persistence Mechanism:** Scheduled Task “PythonUpdate”
- **Execution Method:** PowerShell and Scheduled Task

Investigation Summary (What happened)

On 15 October 2025, a password spraying attack originating from IP address 172.16.0.184 targeted multiple domain accounts on the host FRONTDESK-PC1. Following multiple failed authentication attempts, the attacker successfully authenticated using valid credentials belonging to the user Ryan.Adams.

After gaining access, the attacker executed a malicious payload (python.exe) located in the user’s Music directory. Immediately after execution, the payload initiated outbound network connections to the external IP address 157.245.46.190 over port 9999, establishing a command-and-control (C2) communication channel.

Sustained communication with this external infrastructure indicates active remote control of the compromised system, which aligns with the reported symptoms of unauthorized mouse movement.

To maintain persistence, the attacker created a scheduled task named “PythonUpdate,” configured to execute the malicious payload upon system startup with SYSTEM-level privileges.

Based on the available telemetry, no evidence of lateral movement or further compromise was identified.

Who, What, When, Where, Why, How

Who

- Victim User: Ryan.Adams
- Affected Host: FRONTDESK-PC1
- Attacker Source IP: 172.16.0.184
- Malicious Infrastructure: 157.245.46.190

What

A password spraying attack resulted in the compromise of a user account. The attacker executed a malicious payload, established command-and-control communication, and created persistence via a scheduled task.

When

- Password spraying: 12:51:44 UTC
- Successful login: 12:55:17 UTC
- Payload execution: 13:00:33 UTC
- C2 communication: 13:00:34 UTC onward
- Persistence creation: 13:04:59 UTC

No further activity was observed after 13:06:40 UTC.

Where

The activity occurred on the host FRONTDESK-PC1 within the user's directory:

`C:\Users\Ryan.Adams\Music`

Why

The compromise likely occurred due to weak or reused credentials, making the account susceptible to password spraying attacks. The attacker's objective was to gain unauthorized access and establish persistent remote control over the system.

How

1. The attacker performed a password spraying attack against multiple accounts.
2. The Ryan.Adams account was successfully compromised.
3. The attacker authenticated via network logon.
4. A malicious file (python.exe) was executed from a user directory.
5. The malware established a C2 connection to 157.245.46.190 over port 9999.

6. PowerShell and system utilities were used for post-exploitation.
7. Persistence was established using a scheduled task executing at startup under SYSTEM privileges.

Attack Timeline

Time (UTC)	Event
12:51–12:55	Password spraying activity (multiple failed logins)
12:55:17	Successful authentication (Ryan.Adams)
13:00:33	Malicious python.exe executed
13:00:34	Command-and-control communication initiated
13:04:59	Scheduled task persistence created

MITRE ATT&CK Mapping

- **Initial Access:** T1110 – Brute Force
- **Execution:** T1059 – Command Execution
- **Persistence:** T1053.005 – Scheduled Task
- **Command and Control:** T1071 – Application Layer Protocol

Recommendations

Immediate Actions

- Isolate the affected host from the network.
- Reset passwords for all potentially impacted accounts.
- Remove the malicious file and associated scheduled task.

Threat Hunting

- Search for IP addresses 172.16.0.184 and 157.245.46.190 across the environment.
- Identify additional hosts communicating with the malicious infrastructure.

Prevention Measures

- Implement account lockout policies.
- Enforce multi-factor authentication (MFA).
- Enhance SIEM detection rules for authentication anomalies.
- Block malicious IP addresses at network boundaries.

Long-Term Actions

- Conduct a full forensic investigation of the affected host.
- Consider reimaging the compromised system.
- Improve user awareness and security training.